

A SYNOPSIS ON

OSCloudSync – Automated Google Drive Backup System

**Submitted in partial fulfilment of the requirement for the award of the
degree of**

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

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CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the Synopsis entitled **"OSCloudSync – Automated Google Drive Backup System"** in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science & Engineering of the Graphic Era Hill University, Bhimtal campus and shall be carried out by the undersigned under the supervision of **Mr. Anubhav Bewerwal**, Department of Computer Science & Engineering, Graphic Era Hill University, Bhimtal.

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Internal Evaluation (By DPRC Committee)

Status of the Synopsis: Accepted / Rejected

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Chapter 1 : Introduction

OSCloudSync is an advanced automated file backup system designed to ensure seamless synchronization of files with Google Drive. The system continuously monitors a designated folder, detects any modifications (such as file additions, deletions, or updates), and synchronizes these changes with Google Drive in real time or at scheduled intervals.

To achieve this, OSCloudSync leverages several core Operating System (OS) concepts, including:

File System Management – Efficient handling of files and directories, ensuring smooth operations for detecting, reading, and writing changes.

Process Management – Scheduling multiple backup tasks efficiently, preventing conflicts and optimizing CPU utilization.

Concurrency – Managing multiple file operations simultaneously without data corruption.

Memory Management – Efficiently utilizing system memory to store temporary backup data and reduce unnecessary memory overhead.

I/O Management – Handling input/output operations efficiently to avoid excessive disk or network usage.

Networking – Securely transmitting files to Google Drive while minimizing bandwidth consumption.

By combining these OS concepts, OSCloudSync ensures an efficient, reliable, and user-friendly automated backup solution with minimal user intervention.

These functionalities align with modern cloud backup frameworks [1, 2]

Expanded Problem Statement

Challenges in Data Management and Backup

Data loss can occur due to various reasons, including:

1. **Hardware Failures** – Hard drive crashes, SSD failures, or power outages can lead to permanent data loss.
2. **Accidental Deletions** – Users may mistakenly delete important files without an easy way to recover them.
3. **Software Corruption** – Malware attacks, operating system crashes, or software bugs can cause data corruption.
4. **Inefficient Manual Backups** – Traditional manual backups are time-consuming, prone to human error, and often forgotten.
5. **Resource-Heavy Backup Solutions** – Some backup tools consume excessive system resources, slowing down other processes.
6. **Lack of Effective Failure Recovery** – Many backup solutions fail to recover from interruptions (e.g., network failures, abrupt shutdowns).

How OSCloudSync Solves These Challenges

To overcome these issues, OSCloudSync integrates intelligent backup strategies using key OS functionalities to provide a fast, reliable, and resource-efficient backup system. It incorporates:

1. **Real-time File Monitoring**
 - a. Continuously observes a designated folder using file system event listeners.
 - b. Detects changes (e.g., file creation, deletion, modification) instantly.
 - c. Initiates synchronization to Google Drive without user intervention.
2. **Advanced Process Scheduling**
 - a. Process management to schedule backup operations efficiently.
 - b. Ensures smooth operation without overloading the system.
3. **Failure Recovery Using Priority Queues**
 - a. Implements intelligent failure handling mechanisms.
 - b. Uses priority queues to track and retry failed backup operations.
 - c. Recovers from network failures, interrupted processes, or system crashes.
4. **Storage Optimization Techniques**
 - a. Utilizes incremental backups (only changed files are uploaded, reducing bandwidth and storage usage).
 - b. Supports deduplication to avoid storing duplicate files unnecessarily.
 - c. Compresses files before backup to minimize storage consumption on Google Drive.

Chapter 2: Background/Literature Survey

Literature Review

1. Evolution of Cloud Storage and Automated Backup Systems

Cloud storage has transformed data management by offering scalable, cost-effective, and accessible backup solutions [1]. Traditional manual backups were prone to human error, requiring users to remember periodic file transfers. Automated backup systems address these challenges by ensuring continuous data synchronization, reducing the risk of data loss due to hardware failures, accidental deletions, or system crashes.

Research studies emphasize the importance of key OS concepts—such as real-time file monitoring, process scheduling, and failure recovery—to enhance backup efficiency. OSCloudSync incorporates these advancements, offering a dynamic, self-sustaining backup solution that optimizes resource utilization while maintaining data integrity.

2. Key Components of an Efficient Backup System

A. Real-Time Monitoring & Event-Driven Synchronization

Studies highlight the advantages of event-driven file monitoring over traditional periodic backups [1,2]. Real-time monitoring systems use file system event listeners (e.g., `inotify` for Linux, `ReadDirectoryChangesW` for Windows) to detect file modifications instantly.

B. Process Scheduling and Resource Optimization

Efficient process management is critical for backup software, as uncontrolled file operations can cause CPU bottlenecks high disk usage.

C. Failure Recovery and Intelligent Rollback

Backup failures due to network disconnections, system crashes, or storage limits pose a major challenge. Research suggests that intelligent failure recovery mechanisms using priority queues and checkpointing can significantly improve backup reliability [2].

◇ How OSCloudSync Enhances Failure Recovery:

- Failed sync operations are automatically retried in a priority queue.
- The system resumes from the last successful state, avoiding redundant uploads.
- Checksums and integrity verification ensure data consistency between local and cloud storage.

D. Storage Optimization & Bandwidth Efficiency

Cloud storage costs can escalate with unoptimized backups. Research supports techniques such as incremental backups, deduplication, and compression to minimize storage use and bandwidth consumption.

3. Comparison with Existing Backup Software

Several commercial and open-source backup tools exist, but they have limitations in process efficiency, failure handling, and storage management.

Feature	Traditional Backup Tools	Advanced Backup Systems	OSCloudSync
Real-Time Monitoring	Periodic scans	Event-driven but limited	Instant sync with event-driven monitoring
Process Scheduling	Single-threaded	Multi-threaded but static	Adaptive scheduling with CPU load balancing
Failure Recovery	No automatic retries	Limited rollback	Priority-based retries and integrity checks
Storage Optimization	Full backups only	Incremental backups	Incremental syncing, deduplication, compression
Error Logging & Alerts	Basic logs	Limited alerts	Detailed logs, categorized errors, real-time notifications

Chapter 3 : OBJECTIVES

1. Automated Backup

OSCloudSync ensures seamless file synchronization with minimal user intervention. It monitors designated folders in real time, detects changes, and automatically syncs them with Google Drive. Users can choose between real-time or scheduled backups, ensuring efficiency without manual effort. The backup process runs in the background, optimizing system performance without disruptions.

2. Failure Recovery Mechanism

To prevent data loss, OSCloudSync automatically retries failed sync operations using a priority-based queue. If a failure occurs (due to network issues or system crashes), the system resumes from the last successful checkpoint. It categorizes errors for better troubleshooting and verifies file integrity using checksums to prevent corruption.

3. Storage Optimization

OSCloudSync reduces storage and bandwidth usage through incremental backups, syncing only modified files. It detects and removes duplicate and outdated files based on user-defined retention policies. Compression further minimizes storage consumption, making backups more efficient.

4. Advanced Error Logging

All backup activities and errors are logged for easy tracking and troubleshooting. Logs include timestamps, file statuses, and categorized errors. Users receive real-time alerts for failed syncs, storage issues, or unusual activity. Logs can be exported for deeper analysis and integration with monitoring tools.

Chapter 4 : Hardware and Software Requirements

4.1 Hardware Requirements

S. No	Name of the Hardware	Specification
1.	Processor	Intel i5/i7 or equivalent
2.	RAM	4GB or higher
3.	Storage	50GB (HDD/SSD)

4.2 Software Requirements

S. No	Name of the Software	Specification
1.	Operating System	Windows
2.	Programming language	Python,Java
3.	Cloud API	Google Drive API

Chapter 5: Possible Approach/Algorithms

Approach

1. Real-time Monitoring

OSCloudSync employs OS-level file system watchers (e.g., ReadDirectoryChangesW for Windows) [3] to detect changes in a designated folder. It listens for events such as file creation, modification, renaming, or deletion, ensuring that updates are captured immediately.

◇ Implementation:

- Uses ReadDirectoryChangesW (Windows), for event-driven monitoring.
- Reduces unnecessary file scanning by synchronizing only modified files.
- Prevents data loss by ensuring changes are recorded in a backup queue before upload.

2. Task Scheduling

A queue-based approach efficiently manages file backup operations by prioritizing tasks based on their time of update.

3. Failure Recovery & Priority Queue

Backup failures due to network disconnections, power outages, or file locks are automatically retried using a priority queue, following best practices in cloud synchronization [1, 3]

◇ Recovery Strategy:

- Failed tasks are logged and added to a priority queue for automatic re-execution.
- Retries are prioritized based on failure type:
 - Critical: System files, important documents.
 - Moderate: Regular user files.
 - Low: Temporary/cache files.
- If repeated failures occur, the system alerts the user with error logs and suggestions for resolution.

4. Automated Cleanup for Storage Optimization

To prevent unnecessary storage usage, OSCloudSync automatically removes unmodified files from system memory after 15 days.

◇ Optimization Techniques:

- Incremental backups ensure that only modified files remain in memory for future syncs.
- Redundant/unchanged files are deleted after the set retention period.
- Users can customize cleanup policies to retain files for longer durations if needed.

Algorithms and Pseudocode

1. File Change Detection Algorithm

```
watch_folder(directory):  
    while True:  
        changes = detect_changes(directory)  
        if changes:  
            log_changes(changes)  
            add_to_queue(changes)
```

2. Priority Queue for Failed Processes

```
handle_failed_sync(file):  
    rollback(file)  
    add_to_priority_queue(file)  
    retry_sync()
```

3. Automated Deletion Algorithm

```
delete_old_files(directory):  
    for file in directory:  
        if last_modified(file) > 15_days:  
            delete(file)
```

References

1. John et al., "Cloud Storage and File Synchronization Techniques," IEEE Journal, 2020.
2. Smith & Lee, "Automated Backup Systems: Challenges and Solutions," Computer Science Review, 2019.
3. Google Drive API Documentation , Available at: <https://developers.google.com/drive>.